A. Meximum Array

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: binary search, constructive algorithms, greedy, implementation, math, two pointers |
https://codeforces.com/problemset/problem/1628/A

Mihai has just learned about the MEX concept and since he liked it so much, he decided to use it right away.

Given an array a of n non-negative integers, Mihai wants to create **a new array *b*** that is formed in the following way:

While a is not empty:

- Choose an integer k ($1 \le k \le |a|$).
- Append the MEX of the first k numbers of the array a to the end of array b and erase them from the array a, shifting the positions of the remaining numbers in a.

But, since Mihai loves big arrays as much as the MEX concept, he wants the new array b to be the **lexicographically maximum**. So, Mihai asks you to tell him what the maximum array b that can be created by constructing the array optimally is.

An array x is lexicographically greater than an array y if in the first position where x and y differ x_i gt y_i or if $|x|$ gt $|y|$ and y is a prefix of x (where $|x|$ denotes the size of the array x).

The **MEX** of a set of non-negative integers is the minimal non-negative integer such that it is not in the set. For example, **MEX**({1, 2, 3}) = 0 and **MEX**({0, 1, 2, 4, 5}) = 3.

Input

The first line of the input contains a single integer t ($1 \le t \le 100$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the number of elements in the array a.

The second line of each test case contains n **non-negative** integers $a_1, \ldots, a_n$ ($0 \le a_i \le n$), where a_i is the i-th integer from the array a.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case print m — the length of the maximum array b Mihai can create, followed by m integers denoting the elements of the array b.

Example

Input

```
6
5
1 0 2 0 3
8
2 2 3 4 0 1 2 0
1
1
5
0 1 2 3 4
4
0 1 1 0
10
0 0 2 1 1 1 0 0 1 1
```

Output

```
1
4
2
5 1
1
0
1
5
2
2 2
4
3 2 2 0
```

Note

In the first test case, the lexicographically maximum array *b* is obtained by selecting *k*=5, resulting in the *MEX* of the whole array *a*. It is lexicographically maximum because an array starting with a smaller number than 4 is lexicographically smaller, and choosing a *k* lt 5 would result in an array starting with a number smaller than 4.

In the second test case, there are two ways to obtain the maximum array: first selecting *k*=6, then *k*=2, or first selecting *k*=7 and then *k*=1.